HVAC System Optimization Report

Ottawa Commercial Building Case Study

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HVAC Optimization Project

Realistic TMYx Weather • Part-Load Curves • Monte Carlo Analysis

Executive Summary

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Baseline Performance
    Annual Energy Consumption : 636,413 kWh
    Annual Operating Cost      : $76,369.55

Best Scenario — Combined Measures
    Annual Energy              : 520,613 kWh
    Energy Reduction           : 115,800 kWh (18.2%)
    Cost Savings               : $13,896.02
    Payback Period             : 3.6 years
    CO₂ Reduction              : 57.9 tonnes

Parametric Optimization (SciPy)
    Additional Annual Savings  : $-8,969
    Minimum Annual Cost Found  : $53,504.14

Monte Carlo Analysis (5,000 runs)
    Mean ROI                   : 3.8 years
    95% Confidence Interval    : 2.0 — 6.8 years
    Probability of ROI < 8 yrs : 99.0%
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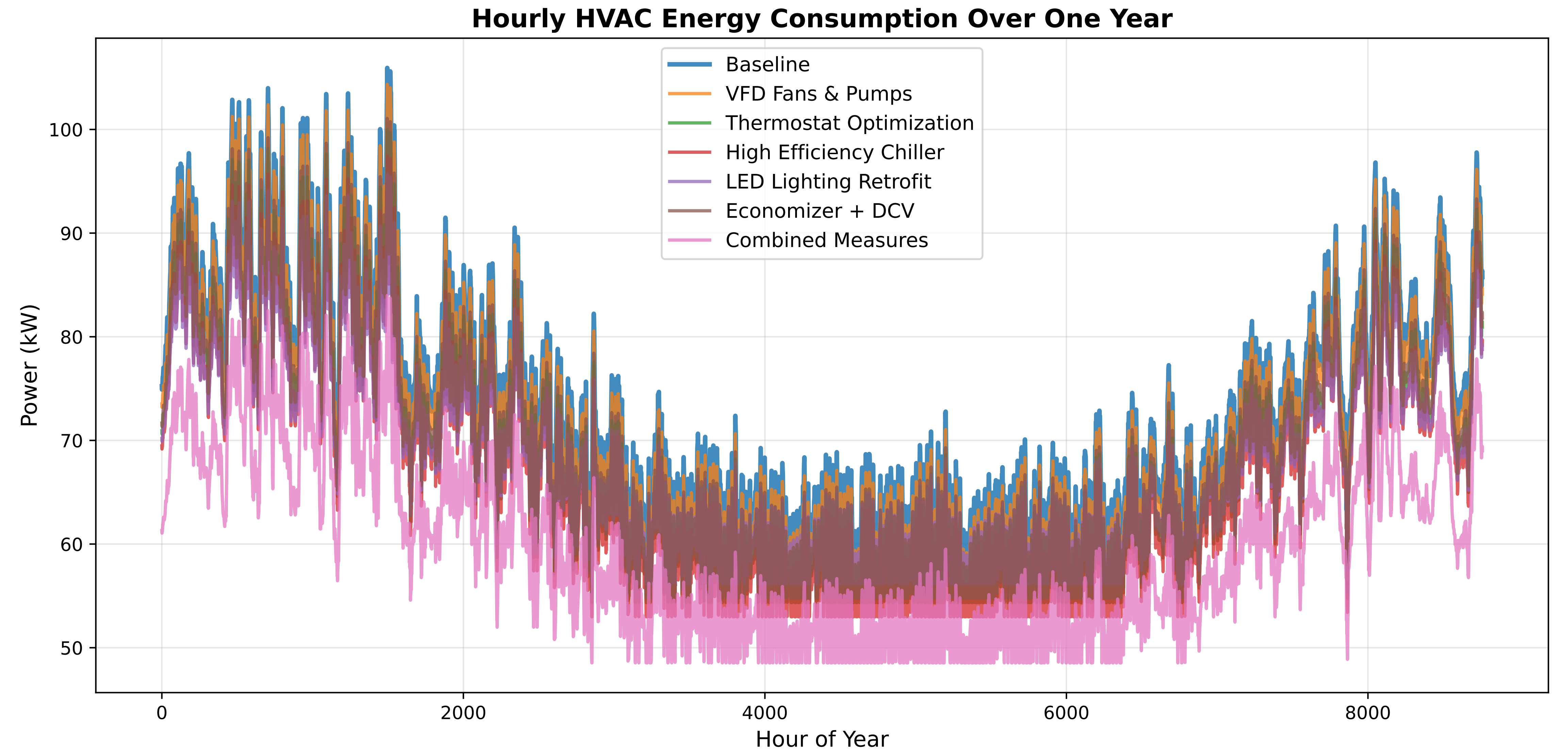
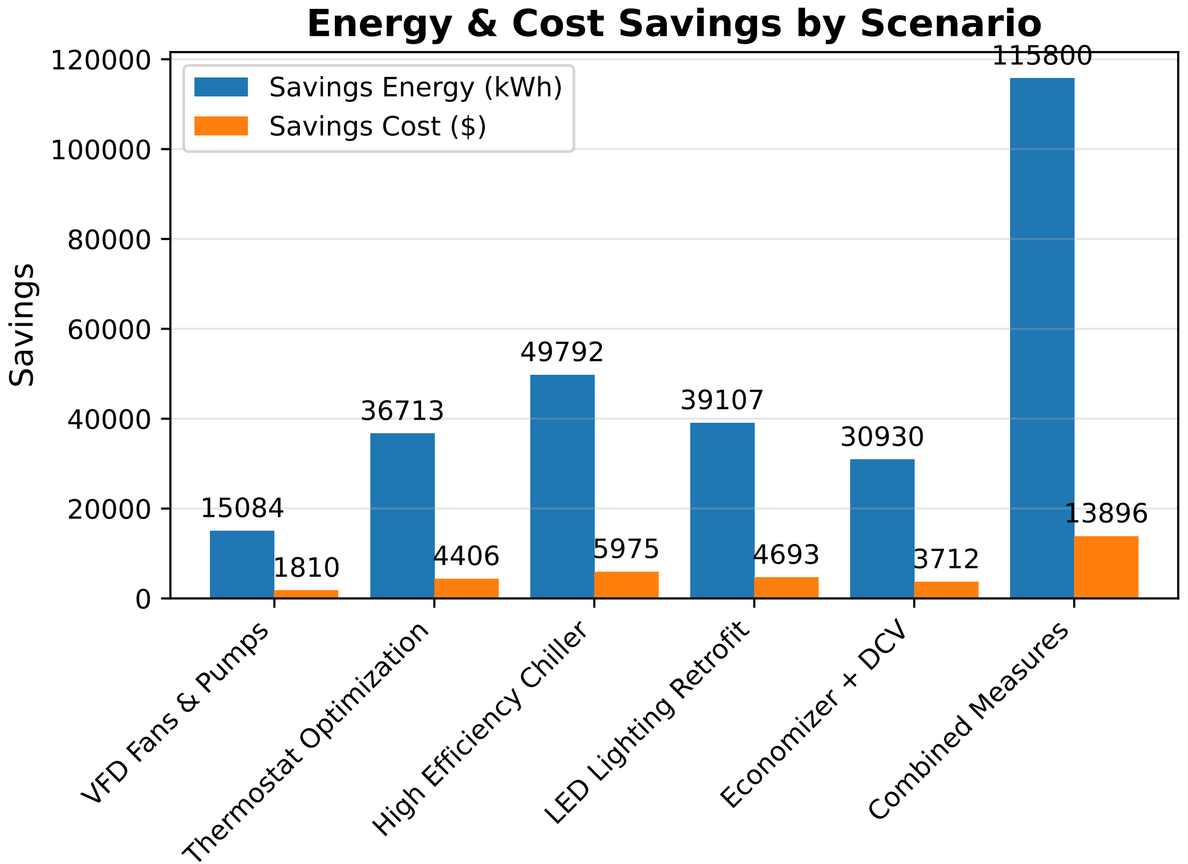


Hourly HVAC Energy Consumption Over One Year
Baseline
VFD Fans & Pumps
Thermostat Optimization
High Efficiency Chiller
LED Lighting Retrofit
Economizer + DCV
Combined Measures
Power (kW)
Hour of Year
0
2000
4000
6000
8000
50
60
70
80
90
100



Energy & Cost Savings by Scenario
Savings
0
20000
40000
60000
80000
100000
120000
Savings Energy (kWh)
Savings Cost ($)
15084
1810
36713
4406
49792
5975
39107
4693
30930
3712
115800
13896
VFD Fans & Pumps
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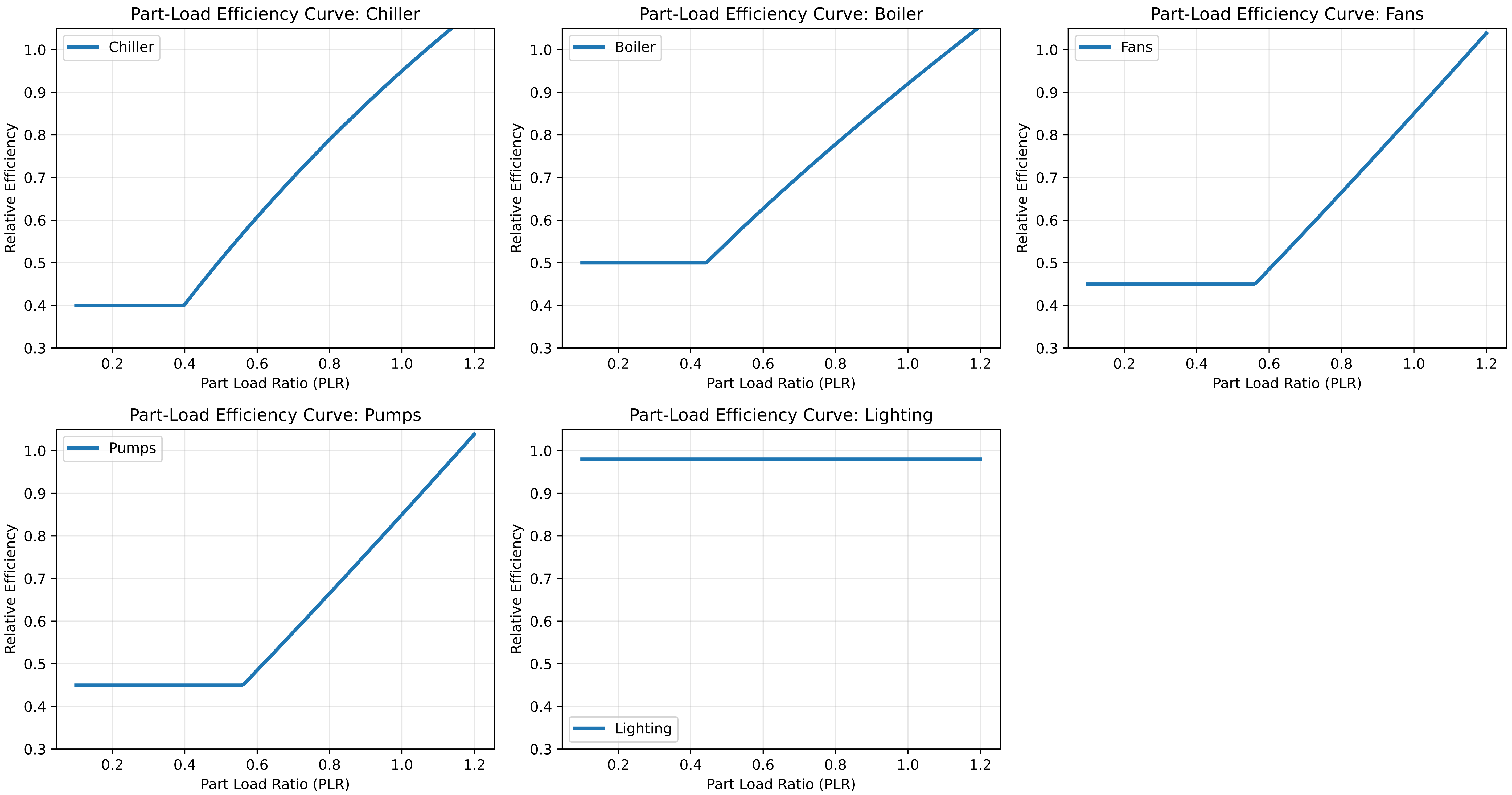
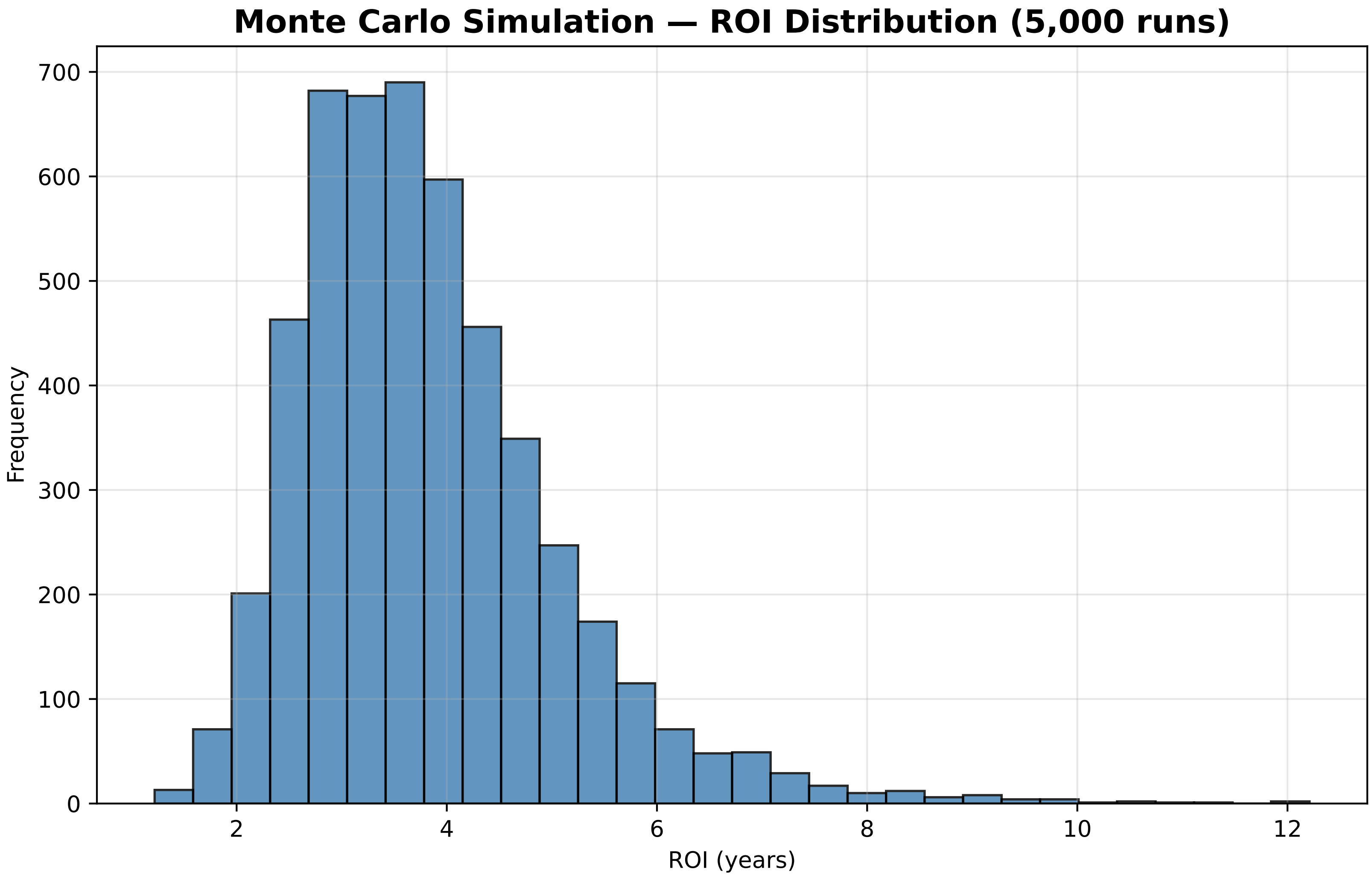
Combined Measures



Part-Load Efficiency Curve: Chiller
Chiller
Part-Load Efficiency Curve: Boiler
Boiler
Part-Load Efficiency Curve: Fans
Fans
Part-Load Efficiency Curve: Pumps
Pumps
Part-Load Efficiency Curve: Lighting
Lighting
Relative Efficiency
Part Load Ratio (PLR)


Monte Carlo Simulation — ROI Distribution (5,000 runs)
Frequency
0
100
200
300
400
500
600
700
2
4
6
8
10
12
ROI (years)

Optimal Configuration from Parametric Optimization

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Recommended Upgraded Component Ratings:

    Chiller      :  22.5 kW   ( 25.0% reduction)
    Boiler       :  12.0 kW   ( 20.0% reduction)
    Fans         :   3.2 kW   ( 36.0% reduction)
    Pumps        :   3.5 kW   ( 30.0% reduction)
    Lighting     :   3.2 kW   ( 60.0% reduction)
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